

prgm 1:

Step 1: Initialize Repository Create a new project folder and initialize Git:

```
mkdir devops-lab1
```

```
cd devops-lab1
```

```
git init
```

Step 2: Configure Git User Set global username and email:

```
git config --global user.name "StudentName"
```

```
git config --global user.email "student@email.com"
```

Step 3: Create and Commit File echo "DevOps Lab" > file1.txt

```
git add .
```

```
git commit -m "Initial Commit"
```

Step 4: Create Branch and Modify File

```
git branch feature1
```

```
git checkout feature1
```

```
echo "Feature Work" >> file1.txt
```

```
git commit -am "Feature Update"
```

Step 5: Merge Branch

```
git checkout main
```

```
git merge feature1
```

Step 6: View History and Changes

```
git log
```

```
git diff
```

prgm2

Step 2: Clone Repository to Local System

```
git clone https://github.com/username/devops-lab2.git
```

```
cd devops-lab2
```

Step 3: Create File and Commit

```
echo "GitHub Lab" > index.html
```

```
git add .
```

```
git commit -m "Added HTML file"
```

Step 4: Push Code to GitHub

```
git push origin main
```

Step 5: Pull Changes from GitHub

```
git pull origin main
```

Step 6: Create Pull Request (Using UI) ➤ Go to GitHub repository ➤ Create new branch ➤

Make changes ➤ Click Compare & Pull Request ➤ Submit PR

prgm3:

Step 2: Clone Repository

```
git clone https://gitlab.com/username/devops-lab3.git
```

```
cd devops-lab3
```

Step 3: Create File and Commit

```
echo "GitLab Project" > app.js
```

```
git add .
```

```
git commit -m "Initial GitLab Commit"
```

Step 4: Push Code to GitLab

```
git push origin main
```

Step 5: Explore GitLab Features ➤ Navigate to Repository → Files ➤ Open Issues tab → Create a sample issue ➤ Go to CI/CD → Pipelines ➤ View repository graph

prgm 4

Step 2: Clone Repository

```
git clone https://username@bitbucket.org/username/devops-lab4.git
```

```
cd devops-lab4
```

Step 3: Create File and Commit

```
echo "Bitbucket Repository" > readme.md
```

```
git add .
```

```
git commit -m "Initial Commit"
```

Step 4: Push Code to Bitbucket

```
git push origin main
```

Step 5: Demonstrate Branch Permissions (UI) ➤ Go to Repository Settings ➤ Navigate to Branch Permissions ➤ Add rule to restrict main branch ➤ Save changes

prgm5

Step 1: Create Sample Web Project

```
mkdir devops-lab5
```

```
cd devops-lab5
```

```
echo "<h1>CI/CD Deployment Success</h1>" > index.html
```

Initialize Git: git init

```
git add .
```

```
git commit -m "Initial Web Project"
```

Step 2: Push Project to GitHub

```
git remote add origin https://github.com/username/devops-lab5.git
```

```
git push -u origin main
```

prgm 6:

Step 1: Pull and Run Nginx Container

```
docker pull nginx
```

```
docker run -d -p 8081:80 nginx
```

Step 2: Verify Running Container open browser <http://localhost:8081>

Step 3: Create Custom Web Application

```
mkdir devops-lab6
```

```
cd devops-lab6
```

```
echo "<h1>Docker Deployment Success</h1>" > index.html
```

Step 4: Create Dockerfile Create a file named Dockerfile:

```
FROM nginx
```

```
COPY . /usr/share/nginx/html
```

Step 5: Build Docker Image

```
docker build -t devops-web .
```

Step 6: Run Custom Container

```
docker run -d -p 8082:80 devops-web
```

prgm7

Step 1: Create Sample Web Application

```
mkdir devops-lab7
```

```
cd devops-lab7
```

```
echo "<h1>CI/CD with Docker Success</h1>" > index.html
```

Step 2: Create Dockerfile

```
FROM nginx
```

```
COPY . /usr/share/nginx/html
```

Step 3: Push Code to GitHub

```
git init
```

```
git add .
```

```
git commit -m "Initial Commit"
```

```
git remote add origin https://github.com/username/devops-docker.git
```

```
git push -u origin main
```

Step 4: Create Jenkins Pipeline Job ➤ Open Jenkins Dashboard ➤ Click New Item ➤ Select Pipeline ➤ Name: DevOps-Lab7

Step 5: Configure Pipeline Script pipeline {

```
agent any
```

```
stages { stage('Clone') {
```

```
    steps {
```

```
        git 'https://github.com/username/devops-docker.git'
```

```
    } }
```

```
    stage('Build Image') {
```

```
        steps {
```

```
            bat 'docker build -t devops-app .' }
```

```
    stage('Run Container') {
```

```
        steps {
```

```
            bat 'docker run -d -p 8090:80 devops-app' }
```

```
    } }
```

prgm8

Step 1: Create Web Application

```
mkdir devops-lab8
```

```
cd devops-lab8
```

```
echo "<h1>GitHub Actions CI/CD Success</h1>" > index.html
```

Step 2: Initialize Git and Push to GitHub

```
git init
```

```
git add .
```

```
git commit -m "Initial Commit"
```

```
git remote add origin https://github.com/username/devops-lab8.git
```

```
git push -u origin main
```

Step 3: Create Workflow Directory

```
mkdir -p .github/workflows
```

Step 4: Create Workflow File

Path: .github/workflows/deploy.yml

Add the following YAML:

name: Deploy Website

on:

push:

branches: [main]

jobs:

build: runs-on: ubuntu-latest

steps:

- name: Checkout Code

uses: actions/checkout@v3

- name: Deploy Step

run: echo "Website Deployed Successfully"

Step 5: Push Workflow File

git add .

git commit -m "Added GitHub Actions workflow"

git push